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SUMMARY

Computer science student at Purdue University working across software engineering, machine learning, and human-centered research. Undergraduate researcher in three labs (wearable sensing, robotic manipulation, XR security), with industry experience shipping production ML pipelines and backend systems. Leads 20-contributor student software teams; co-founded a game development club that shipped 10 Unity titles.

EDUCATION

Purdue University — College of Science

B.S. in Computer Science (Machine Intelligence & Systems Software Concentration)

West Lafayette, IN

Aug 2024 – May 2027

- **Minor:** Mathematics
- **Relevant Coursework:** Artificial Intelligence, Algorithm Analysis, Data Structures & Algorithms, Systems Programming (C, Assembly), Computer Architecture, Object-Oriented Programming (Java), Competitive Programming (C++), Linear Algebra, Multivariable Calculus

RESEARCH EXPERIENCE

Purdue University — Snowball, CoRAL, and PurSec Labs

Undergraduate Researcher

West Lafayette, IN

Nov 2024 – Present

- **Snowball (Multimodal Wearable Sensing):** Developed the data processing pipeline for smart-earring sensor data (PPG, accelerometer, temperature, audio), including cross-sensor synchronization, preprocessing, cleaning, and dataset generation across 15+ participants and 30+ hours of recordings. Engineered and evaluated features to train ML models for eating detection and chewing-count estimation, reaching 97% F1 and 0.634 MAE.
- **CoRAL (Robot Learning):** Analyzed state-of-the-art deep learning and Vision-Language Model (VLM) architectures for long-horizon robotic manipulation; conducted literature reviews and evaluated VLM reasoning on object-relocation tasks with obstacles across 10 scenarios.
- **PurSec (XR Security & Privacy):** Conducted literature reviews on XR privacy and security, LLM-enabled AR attack vectors, and user/bystander risks. Designed a comprehensive survey and interview study (200+ participants) investigating awareness gaps, willingness to adopt defenses, and novel XR threats; reviewed 20+ papers.

INDUSTRY EXPERIENCE

Reclub — Sports Technology Startup

Software Engineer Intern

Ho Chi Minh City, VN

May 2025 – Aug 2025

- Designed and deployed an automated video segmentation system identifying pickleball rallies from raw match footage, reducing manual editing time from hours to ~5 minutes per match.
- Built and optimized a YOLOv10s-based, motion-aware classification pipeline achieving 96% accuracy while targeting near real-time, on-device inference.
- Implemented an end-to-end ML pipeline — 36K+ frame data labeling, training, and cloud deployment — using Docker, Google Cloud Storage, and Vertex AI.
- Productionized model inference workflows supporting scalable content generation, contributing to increased user engagement and platform adoption.

Amazon Web Services (AWS) Youth Experience Program

Data Analytics Intern

Remote

Jul 2023 – Aug 2023

- Developed Python-based data processing pipelines (Pandas, NumPy, scikit-learn) to analyze wildfire risk, predicting 7 forest cover types from 54 cartographic features on the 581K-sample UCI Covtype dataset.
- Evaluated 5 model families (Logistic Regression, Decision Trees, Random Forests, Neural Networks, KNN) to identify optimal features and improve predictive accuracy; recognized as a top-5 intern project.

BuyMed — \$64M Healthcare Technology Platform Startup

Retail Operations Intern

Ho Chi Minh City, VN

Jul 2023 – Aug 2023

- Reviewed legal contracts with local pharmacies, ensuring compliance and alignment with development operations.
- Conducted market research on competitors and the healthcare landscape; analyzed operational data across 3 pharmacy partners and authored standard operating procedures to improve efficiency.

UNIVERSITY EXPERIENCE

Purdue University — College of Liberal Arts

Web Developer

West Lafayette, IN

Jan 2026 – Present

- Maintain and extend university web services by developing reusable templates and components using HTML, CSS, JavaScript, and Apache Velocity.
- Resolve support tickets involving content updates, accessibility improvements, and legacy system maintenance.

Purdue University

COSINE Mathematics Tutor

West Lafayette, IN

Jan 2025 – May 2025

- Supported 50+ undergraduate students across Precalculus and Calculus I–II (MA 15300–16600), improving exam performance by reinforcing conceptual understanding and adapting to diverse learning styles.

LEADERSHIP & TEACHING

Purdue ACM SIGAPP

President & Technical Lead

West Lafayette, IN

Aug 2025 – Present

- Led backend development for two large-scale student-run software projects with ~20 contributors each, coordinating architecture decisions, sprint planning, and code reviews.
- Designed REST APIs, database schemas, and authentication flows for *BoilerBuzz*, a cross-platform mobile app backed by Node.js and PostgreSQL.
- Architected a FastAPI-based grant management platform for Purdue PGSG supporting submission, review, approval, and reimbursement workflows; implemented role-based access control with Keycloak for secure multi-role, multi-stage approval pipelines.

SSIS Programming Club

Co-Leader

Ho Chi Minh City, VN

Aug 2023 – May 2024

- Taught 15 students data structures, trees, and algorithms in C++ and Python to prepare for USACO competitions; led weekly timed practice sessions and facilitated student presentations on advanced CS topics.

Unity Impact — SSIS Video Game Development Club

Co-founder & President

Ho Chi Minh City, VN

Aug 2022 – Dec 2023

- Designed curriculum, organized a public game exhibition (30+ visitors), and led a winter bootcamp for 21 participants from Vietnam, Australia, and the United States.
- Guided development of 10 games across puzzle to adventure genres using Unity and C#, incorporating 2D sprites and 3D models.

PROJECTS

Boiler-Gram

Multi-threaded Java Client-Server Application

West Lafayette, IN

Oct 2024 – Dec 2024

- Built a multi-threaded client-server system supporting user authentication, messaging, and persistent storage; designed file-backed data models, custom exception handling, and comprehensive JUnit tests.

Roadie Rescue — 3rd Place of 35 Teams, Hello World Purdue Hackathon

Real-Time Roadside Assistance Platform (MERN)

West Lafayette, IN

Oct 2024

- Developed a real-time emergency assistance platform with location-based requests and live status updates.

SchoolOfAthens.world — Platform for Youth Debate Perspectives

Jan 2023 – Jun 2023

- Collaborated with 7 international developers on a real-time data visualization platform (Next.js, Firebase); designed interactive UI and engagement-tracking dashboards visualizing live discussions.

SKILLS

- **Languages:** Python, Java, C++, C, JavaScript, TypeScript, SQL, Assembly
- **Backend & Systems:** FastAPI, Node.js, Express, REST APIs, PostgreSQL, MongoDB, Keycloak (RBAC), multi-threading
- **ML & Data:** scikit-learn, Pandas, NumPy, YOLO/object detection, feature engineering, model evaluation, sensor signal processing
- **Cloud & DevOps:** Docker, Google Cloud Platform (GCS, Vertex AI), Git/GitHub
- **Game Development:** Unity, C#, 2D/3D asset integration

HONORS & AWARDS

- 3rd Place — Hello World Purdue Hackathon (2024)
- Top-5 Intern Project — AWS Youth Experience Program (2023)
- 2nd Place — Girls In Tech Code/No-Code Hack, Lick Wilmerding High School (2022)
- Public Award, Most Voted Innovative Idea — Pooleville High School Hackathon (2022)
- 1st Place — SSIS Inclusive Invention Challenge (2020)